INKLOOM AI CHALLENGE · ROUND 3 · THE FINISH

GUARDIAN AI

The Living Claim Ledger

Claims that stay connected to their evidence.

AI-powered content verification, evidence intelligence and claim monitoring.



The three claim statuses tracked by the Living Claim Ledger.

TRY IT NOW · NO LOGIN NEEDED

guardian-ai-smoky-theta.vercel.app/demo

Public interactive demo on sample data. Section 18 is a two-minute evaluator guide.
Code: github.com/Varun-men/Guardian_AI

TEAM

The Bloodline

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HACKATHON

Inkloom AI Challenge

SUBMISSION

Round 3: The Finish

DATE

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INSIDE THIS SUBMISSION

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Start here: open guardian-ai-smoky-theta.vercel.app/demo. No login is needed. Section 18 is a two-minute evaluator guide, and every claim in this report is stated at the level the evidence supports (see Section 15).

AT A GLANCE

11 pipeline stages from draft to tracked claim

3 ledger statuses: Current, Stale, Unverified

0 approvals made by AI. Humans decide, backend enforces

7 claims in the Orbit X1 validation scenario

01 Executive Summary

The idea in one page.

In one sentence: Guardian AI checks every factual claim in marketing content against approved evidence, keeps a human in charge of approval, and remembers each approved claim in a Living Claim Ledger so it can flag the claim when the evidence changes.

Marketing and product teams publish claims about products, pricing, specifications and features. Those claims go stale when product information changes, and a one-time proofread cannot show which source a claim depends on.

Guardian AI is an AI-assisted content verification platform. It extracts the factual claims in a draft, checks them with deterministic rules and against approved evidence, and puts every finding in front of a human reviewer. The backend enforces approval. Approved facts are then kept in the Living Claim Ledger, linked to their evidence and source versions, so the system can tell when a claim needs renewed verification.

Evidence-backed
Claims are checked against passages from approved sources, with source versions and citations kept.

Human-controlled
Reviewers apply or dismiss each finding, and the backend blocks approval while decisions are pending.

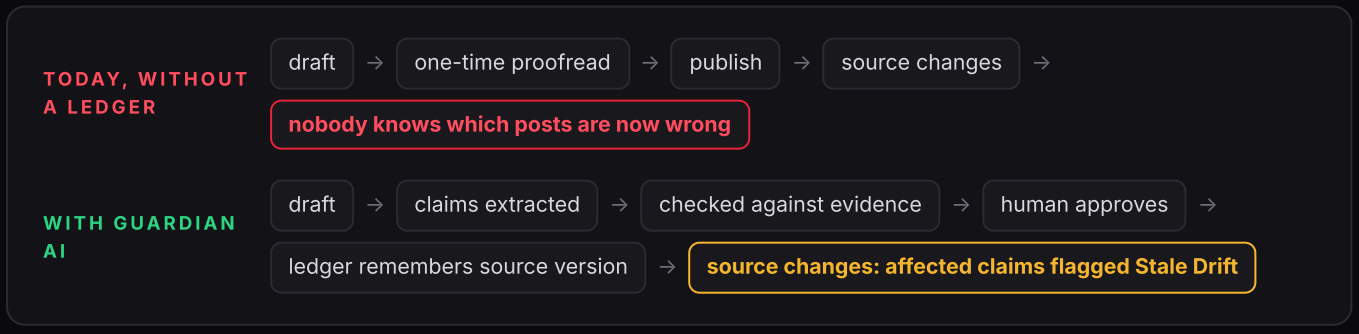
Persistent
The Living Claim Ledger keeps approved facts connected to their evidence and flags Stale Drift.

02 Problem Statement

Content is only as accurate as the sources behind it.

Claims go stale
When product information changes, or content was written from an outdated source, published claims quietly become wrong.

No lasting link to evidence
Proofreading and manual verification take time, and they leave no persistent connection between a published claim and the evidence that supports it.



Guardian AI's answer: extract factual claims, check them against approved evidence, support human review, and keep a persistent record of verified claims in a Living Claim Ledger.

03 Our Solution

Ten capabilities, one pipeline.

01 Analyses marketing content and extracts factual claims.	02 Checks content using deterministic rules.
03 Retrieves supporting evidence from approved sources.	04 Verifies claims against relevant evidence.
05 Identifies supported, contradicted and unverified claims.	06 Provides suggested corrections and evidence references.
07 Lets human reviewers apply or dismiss findings.	08 Controls final approval through backend-enforced review gates.
09 Maintains approved facts and their source relationships in a Living Claim Ledger.	10 Tracks claim status when evidence or source versions change.

The central concept: a claim should not simply be checked once. It should remain connected to the evidence that supports it.

04 What Makes Guardian AI Different

Most content-checking tools review a draft once and move on.

A COMMON APPROACH	GUARDIAN AI
Checks content once, before it is published	The Living Claim Ledger keeps approved claims connected to evidence and source versions, and flags Stale Drift when that evidence changes.
Gives feedback on a document as a whole	Extracts individual factual claims and verifies each one. Severity and verification status are kept separate.
Asks the AI to cite its sources	Validates citations and source-version relationships, and traces each claim to a passage.
Uses an AI model for every check	Combines deterministic rules, AI-assisted extraction and evidence-backed verification.
Treats human review as an afterthought	Persists Apply and Dismiss decisions, and the backend refuses approval while decisions are pending.
Needs credentials to try the product	A public **/demo** with sample data needs no login.

05 Responsible by Design

Four commitments the product is built around.

AI never approves

Approval is a human decision, enforced by the backend. The AI assists; it does not sign off.

Certainty where it exists

Deterministic rules handle rule-based checks, and stay distinct from evidence-backed verification.

Honest about gaps

A claim without sufficient evidence is shown as Unverified instead of being assumed correct.

Demo kept separate

The public demo uses sample data and is separate from authenticated team workflows.

06 Product Features

Seven feature areas. The ledger is the centrepiece.

5 · Living Claim Ledger (the central feature). A persistent registry of approved facts, connected to claims, supporting evidence and source versions, with Current & Synced, Stale Drift and Unverified statuses. Section 08 covers it in detail.

1 AI-Assisted Content Analysis

- Extracts individual factual claims from marketing and product content
- Flags potentially inaccurate statements, with an explanation
- Unique ID for every finding
- Verification status kept separate from severity
- Suggested corrections
- Fallback claim extraction when the primary AI path is unavailable

2 Deterministic Rule-Based Verification

- Rule engine for restricted or problematic wording
- Detects competitor references and pricing inconsistencies
- Flags accessibility issues such as missing image alt text
- Generates structured findings
- Kept distinct from evidence-backed verification

3 Evidence Intelligence & Source Management

- Upload and manage reference documents
- Extract and store relevant evidence passages
- Associate claims with supporting evidence
- Maintain source and source-version relationships
- Validate evidence references and citations
- Trace claim → passage → source version

4 Review Workspace

- Findings with explanations, evidence and suggested corrections
- Apply or Dismiss, with decisions persisted
- Controlled updates to draft content
- Review history
- Backend blocks final approval while decisions remain unresolved

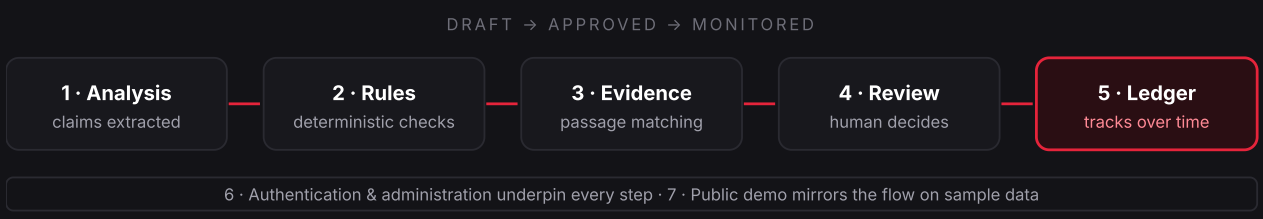
6 Authentication & Administration

- JWT-based authentication for protected team workflows
- Protected application routes
- Source management and administration
- Persistent review and application records
- Backend-enforced approval restrictions

7 Public Interactive Demo

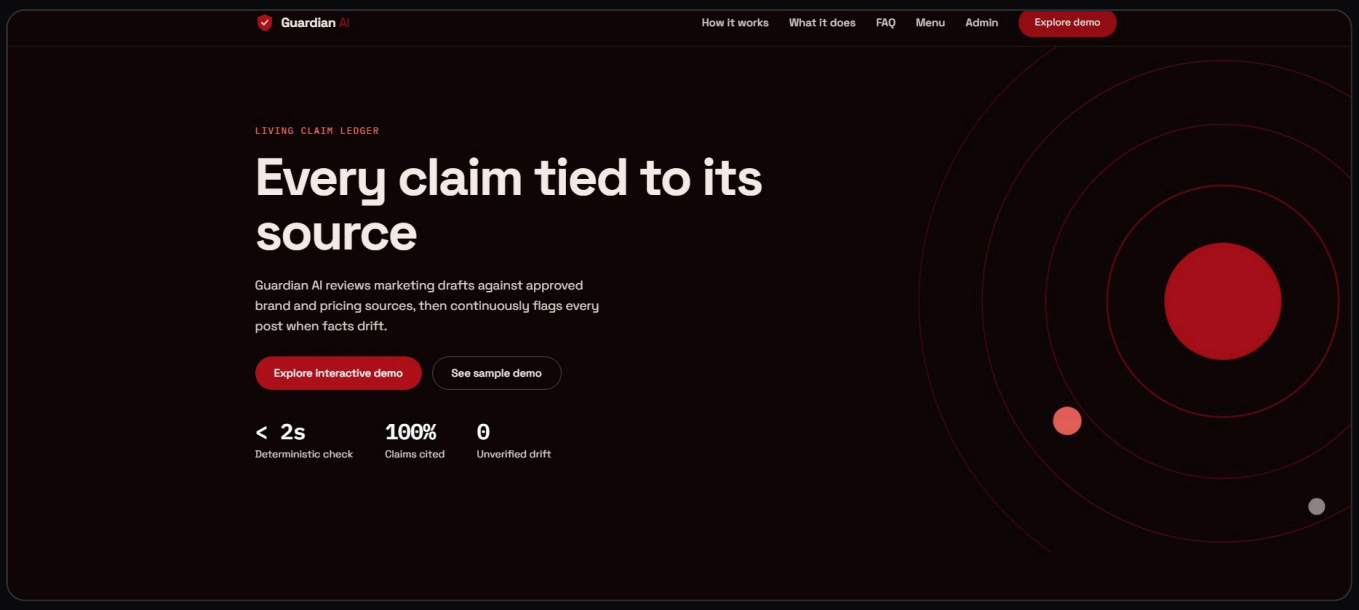
- Public /demo route, no login required
- Uses sample data
- Lets judges follow the product workflow without credentials
- Clearly separate from authenticated team workflows

HOW THE SEVEN AREAS FIT TOGETHER



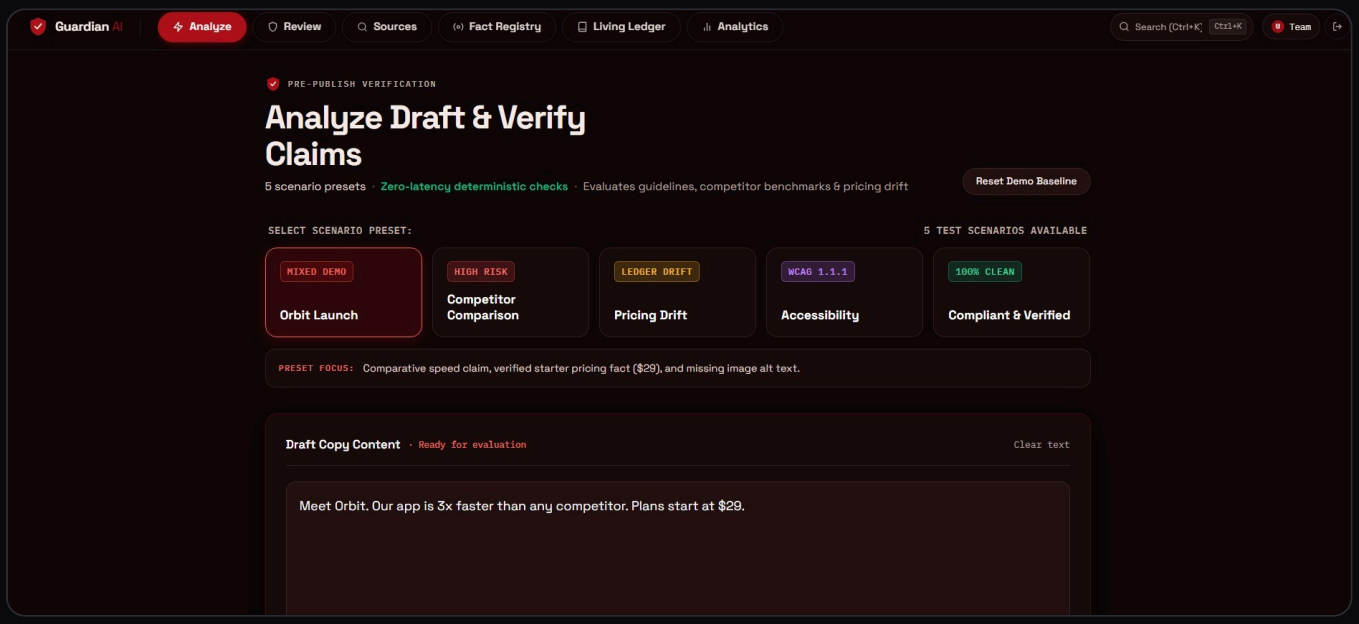
07 Product in Pictures

Screenshots are from the deployed application. Screenshot 4 is an illustrative mockup and is labelled as such.



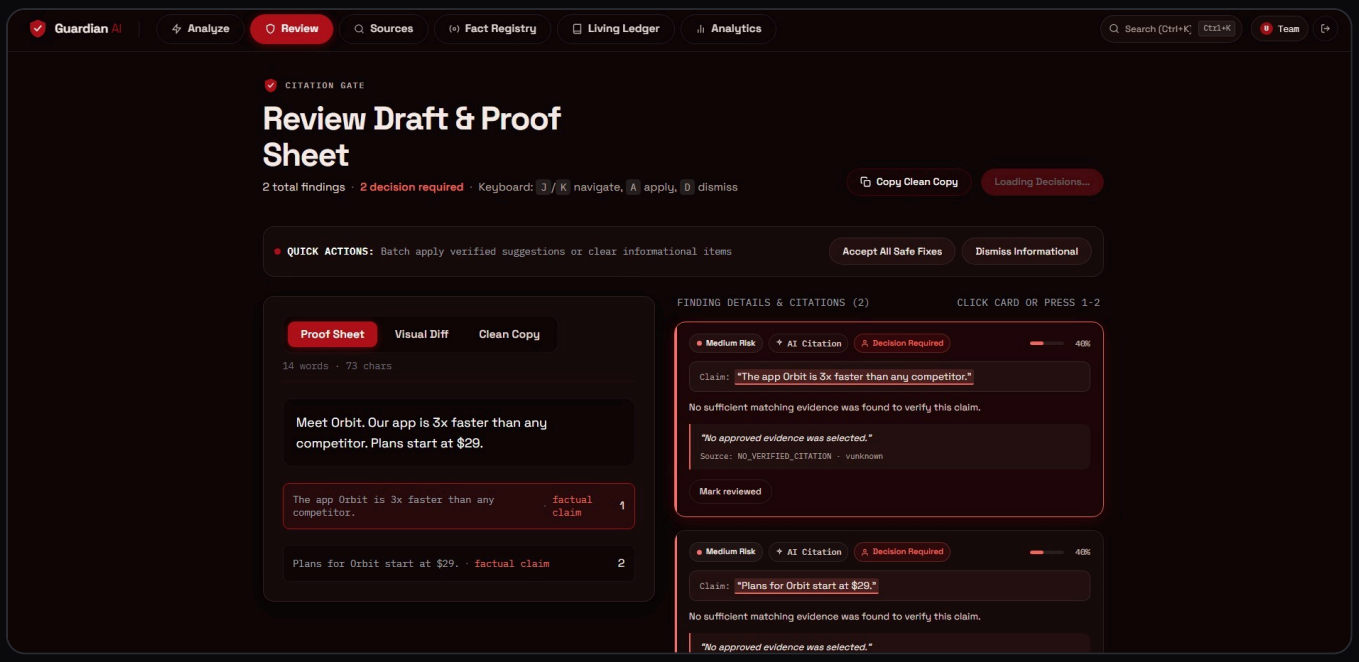
Screenshot 1. The Guardian AI homepage and navigation, including the route to the public demo. Homepage figures are illustrative marketing content, not measured results.

What to notice → the product's promise is on the first screen: every claim tied to its source.



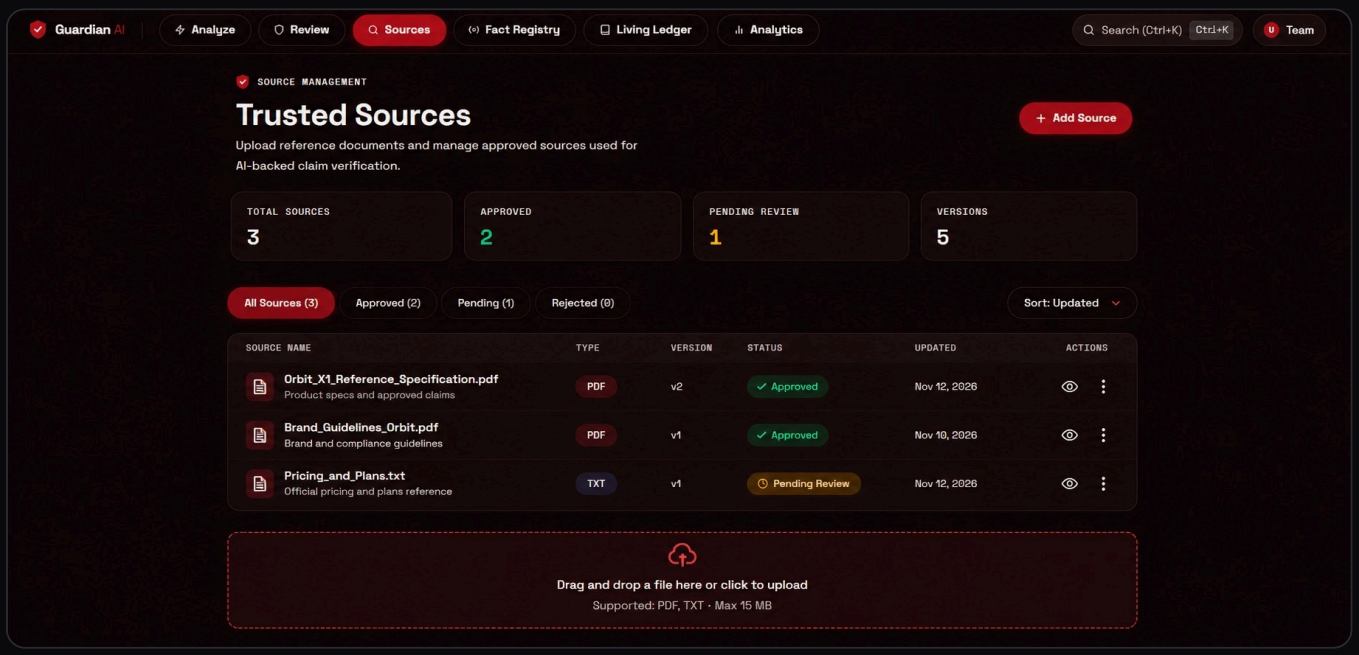
Screenshot 2. Analyze screen: pick a scenario preset or paste a draft, then run verification.

What to notice → five scenario presets (mixed, competitor comparison, pricing drift, accessibility, compliant) let a reviewer exercise each rule path in seconds.



Screenshot 3. The review workspace: each finding shows its claim, evidence status and the Apply and Dismiss decisions.

What to notice → with no approved evidence selected, the claim is shown as unverified and a decision is required. The product never fills the gap with a guess.



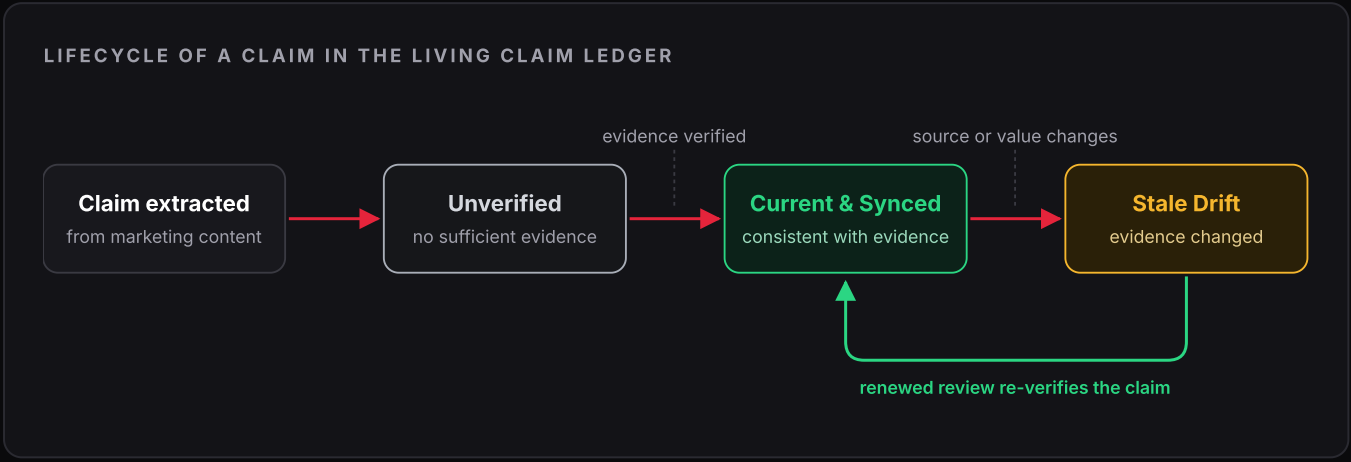
Screenshot 4. Source management: reference documents and their versions **Illustrative UI mockup** not a live upload capture.

What to notice → every source has a version and an approval status. That version is what the ledger links claims to.

08 Living Claim Ledger

The core innovation of Guardian AI.

The Living Claim Ledger is a persistent registry of approved facts. It connects each fact with the claims that rely on it, the evidence that supports it, and the source version that evidence came from. Because those links persist, a claim stays tied to its evidence long after it was approved.



Current & Synced
The claim remains consistent with its linked evidence.

Stale Drift
The supporting evidence has changed or become outdated, requiring renewed review.

Unverified
The claim does not currently have sufficient verified evidence.

What the ledger holds

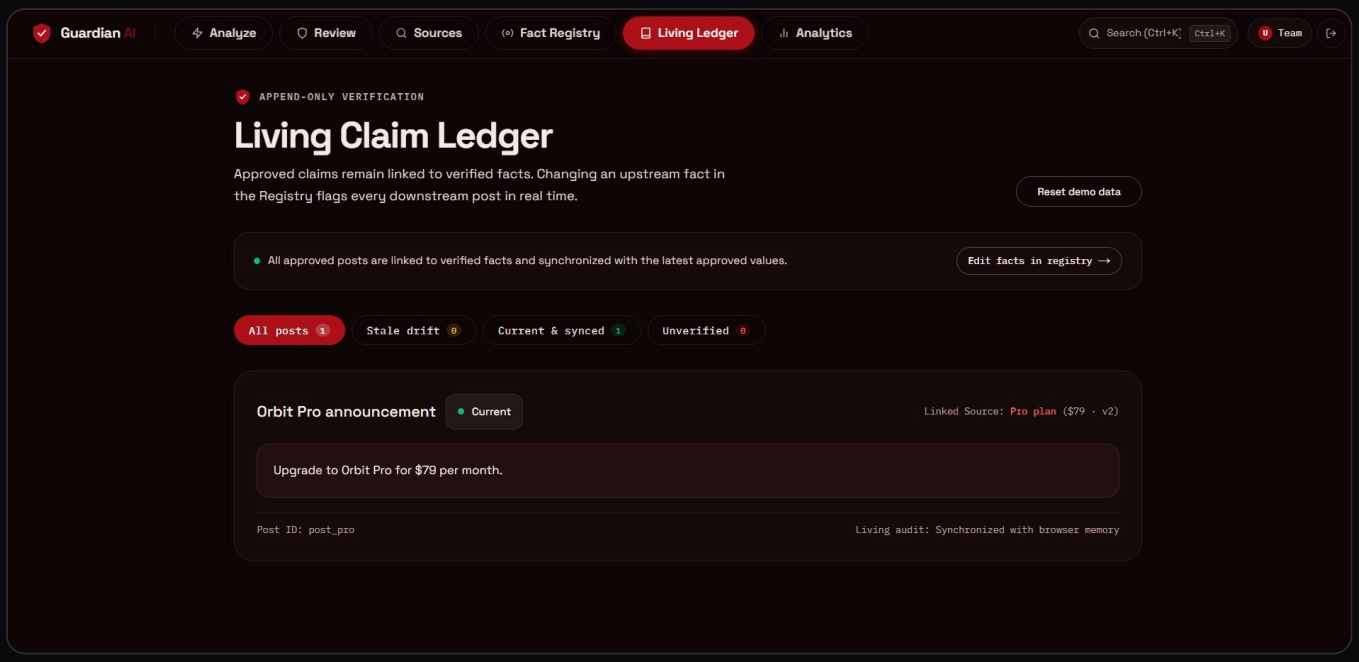
- Persistent fact registry
- Claims linked to approved facts
- Evidence and source-version relationships
- Claim status tracking

What it lets a team do

- Detect outdated or changed supporting evidence
- Identify claims requiring renewed verification
- Use the ledger dashboard and filtering
- Keep records across sessions

How Stale Drift appears

Each ledger entry is linked to a source version. In the screenshot below, the Orbit Pro announcement is linked to the Pro plan at version v2. When a linked source is updated to a new version, or the approved value changes, the claims that depend on it are flagged Stale Drift so a reviewer knows exactly which content needs renewed verification.



Screenshot 5. The Living Claim Ledger dashboard with claim statuses and filtering. Shown from the public demo, where ledger data is held in the browser; in the authenticated application the ledger is stored in PostgreSQL.

Illustrative walk-through

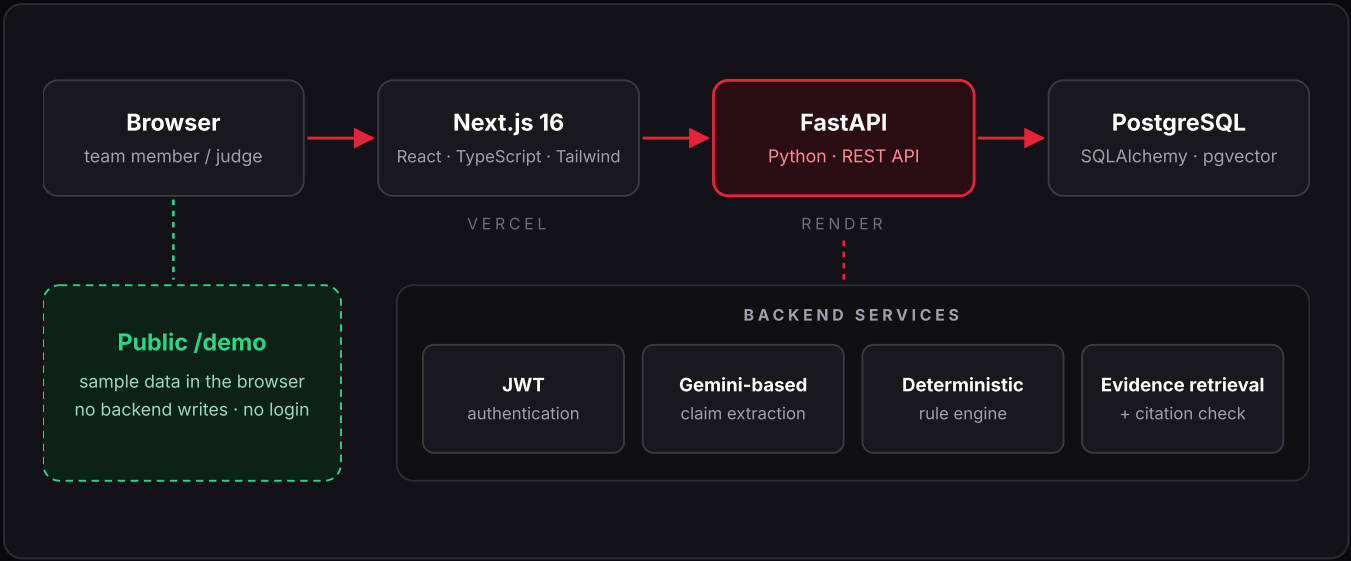
Based on the demo's Orbit Pro entry. It shows the logic of the ledger, not measured behaviour.

STEP	LINKED SOURCE	CLAIM IN THE LEDGER	STATUS
1 · Approved today	Pro plan · $79 · v2	"Upgrade to Orbit Pro for $79 per month."	Current & Synced
2 · Source updated to v3	Pro plan · changed value · v3	Same published text, now out of step with its source	Stale Drift
3 · Reviewer re-verifies	Pro plan · v3	Copy corrected, approved against v3	Current & Synced

Value: teams no longer rely on memory or on re-reading old content. The ledger shows which claims depend on which evidence, and which ones need attention.

09 Technical Architecture

Next.js on Vercel, FastAPI on Render, PostgreSQL with pgvector.



Team workflows run through the authenticated frontend and backend. The public /demo is separate. AI assists extraction; rules and evidence-backed checks add verification; human review stays in the approval path.

LAYER	TECHNOLOGY	RESPONSIBILITY
Frontend	Next.js 16, React, TypeScript, Tailwind CSS	Marketing website, content analysis interface, review workspace, source management, admin and analytics views, Living Claim Ledger dashboard, public interactive demo
Backend	Python, FastAPI, REST API	Content analysis, claim extraction, rule-based checks, evidence retrieval, claim verification, review decisions, approval workflows, source management, ledger operations
AI and verification	Gemini-based AI, deterministic rule engine	Gemini-based claim extraction; rules for rule-based checks; passage-level evidence matching; citation and source-version validation; structured verification results
Database	PostgreSQL, SQLAlchemy, pgvector	Persistent storage of application records, source information, evidence passages, findings, decisions, facts and their relationships
Auth and deployment	JWT, Vercel, Render	JWT-protected team workflows; frontend on Vercel; backend on Render

10 End-to-End Workflow

From submitted content to a tracked claim.

01	USER	**Content submission**	User submits marketing or product content
02	AI	**AI-assisted claim extraction**	Gemini-based extraction; fallback path if unavailable
03	RULES	**Deterministic rule checks**	Restricted wording, competitors, pricing, missing alt text
04	EVIDENCE	**Evidence retrieval**	Passage-level matching against approved sources
05	EVIDENCE	**Evidence-based claim verification**	Supported, contradicted or unverified, with references
06	SYSTEM	**Structured findings & corrections**	Unique IDs; severity and verification status kept apart
07	HUMAN	**Human review: Apply / Dismiss**	Reviewer decides; decisions are persisted
08	BACKEND	**Backend approval gate**	Server refuses approval while decisions are pending
09	LEDGER	**Approved facts & evidence links**	Facts stored with claims, passages and source versions
10	LEDGER	**Living Claim Ledger**	Persistent registry of facts, claims and statuses
11	LEDGER	**Claim status monitoring**	Flags Stale Drift when evidence or source versions change

The verification pipeline. Highlighted rows are where people and the backend take control; the ledger takes over after approval.

The same flow in eight steps

1 Submit content
A user submits marketing or product content for analysis.

2 Extract claims
The system identifies individual factual claims in the submitted content.

3 Analyse and verify
The rule engine checks content patterns, while evidence retrieval and verification evaluate factual claims against available sources.

4 Review findings
The reviewer sees explanations, evidence references, severity, verification status and suggested corrections.

5 Make human decisions
The reviewer applies or dismisses findings. Decisions are saved.

6 Approve content
The backend enforces approval restrictions and prevents approval while required review decisions are pending.

7 Maintain the ledger
Approved facts are stored with their associated claims, evidence and source versions.

8 Track changes
When supporting evidence changes, the ledger identifies claims that may need renewed verification.

11 Orbit X1 Validation Scenario

A concrete end-to-end example.

> **Note:** Orbit X1 is a synthetic test product used for validation. It is not a real commercial product specification.

A reference document with approved information for Orbit X1 is uploaded, and a marketing draft containing seven factual claims is submitted. The system extracts the seven claims and compares each with the approved information.

ATTRIBUTE	APPROVED INFORMATION
Battery	Up to 10 days
Display	1.8-inch AMOLED
GPS	Built-in GPS supported
Water resistance	Up to 50 metres
Heart rate	Continuous monitoring supported
Price	₹5,499
Colours	Black and Blue

4
claims expected to be supported

3
claims expected to be contradicted

Test draft and expected verification

#	CLAIM IN THE DRAFT	EXPECTED RESULT	SUGGESTED CORRECTION
1	Battery life of 14 days	Contradicted approved: up to 10 days	**10 days**
2	1.8-inch AMOLED display	Supported	None needed
3	Built-in GPS	Supported	None needed
4	Water resistance up to 50 metres	Supported	None needed
5	Continuous heart-rate monitoring	Supported	None needed
6	Price of ₹4,999	Contradicted approved: ₹5,499	**₹5,499**
7	Available in Black, Blue and Silver	Contradicted Silver not approved	**Black and Blue**

After applying the suggested corrections (battery 10 days, price ₹5,499, colours Black and Blue), all seven claims match the synthetic reference information.

WHAT THE SCENARIO DEMONSTRATES

Claim extraction
Seven factual claims identified in one draft.

Evidence comparison
Each claim compared with the approved reference.

Contradiction detection
Battery, price and colour claims conflict with the reference.

Suggested corrections
Each contradiction comes with a corrected value.

Human review
The reviewer applies or dismisses each correction.

Source-backed verification
After correction, all claims match the approved source.

Scope: it is a demonstration scenario, not a benchmark, and no accuracy percentage is claimed.

12 Testing and Validation

Validation areas covered.

Functional testing

- Content submission and claim extraction
- Rule-based finding generation
- Unique finding ID generation
- Structured finding status and severity
- Suggested correction generation
- Evidence upload and source management
- Passage-level evidence retrieval
- Claim-to-evidence association
- Citation and source-version relationships
- Apply and Dismiss review decisions
- Persistence of review decisions
- Controlled draft updates
- Backend approval restrictions
- Ledger record creation and retrieval
- Claim status tracking
- Stale evidence and claim identification

Integration testing

- Frontend-to-backend API communication
- Backend database connectivity
- Source and evidence persistence
- Review and approval workflow integration
- Ledger and fact relationship validation
- Authentication-protected workflows

Deployment validation

- Frontend deployed on Vercel
- Backend deployed on Render
- API connectivity verified
- Public demo route accessible without login
- Protected team workflows use authentication

End-to-end testing. The complete workflow was exercised from content submission through claim extraction, evidence-based verification, human review, backend-enforced approval and Living Claim Ledger tracking. The Orbit X1 scenario in Section 11 is the concrete validation example.

This report does not state test counts, response times, uptime figures or accuracy percentages.

13 Round 2 Commitments vs Round 3

The three commitments keep their names and intent. Where Round 3 delivered the foundation of a commitment, it says so.

IMPROVEMENT 1 Advanced Evidence Intelligence

ROUND 2 COMMITMENT	ROUND 3 DELIVERY	OUTCOME
• Source versioning • Reliable document retrieval • Citation validation • Outdated and contradictory evidence detection • Traceability between findings, evidence passages and source versions	• Evidence ingestion and passage-level associations • Source and source-version relationships • Citation validation • Claim-to-evidence traceability • Persistent facts linked to supporting evidence • Identification of stale evidence associations	A more reliable evidence-backed verification process in which claims trace to their supporting sources and source versions.

IMPROVEMENT 2 Collaborative Review and Audit System

ROUND 2 COMMITMENT	ROUND 3 DELIVERY	OUTCOME
• Multi-user review workflows • Role-based permissions • Finding assignments • Reviewer activity history • Auditable decision trail	• Authenticated team workflows • JWT-protected access • Persistent Apply and Dismiss decisions • Backend-enforced approval gates • Saved review decisions • Review history and audit records	A controlled human-review workflow with persistent decisions and an auditable approval process. Collaborative review and audit foundation. Advanced assignments and workflow customisation are on the roadmap.

IMPROVEMENT 3 Automated Continuous Claim Monitoring

ROUND 2 COMMITMENT	ROUND 3 DELIVERY	OUTCOME
• Monitor approved content over time • Detect changed claims and outdated evidence • Generate review alerts • Allow teams to reopen affected reviews	• Persistent Living Claim Ledger • Fact-linked claims • Source-version relationships • Current & Synced, Stale Drift and Unverified statuses • Identification of claims affected by changes to supporting evidence	A persistent, source-aware claim tracking foundation that helps teams identify claims requiring renewed verification. Scheduled monitoring, notifications and automatic review reopening are on the roadmap.

14 Additional Improvements

Made during development, beyond the three commitments.

#	IMPROVEMENT	STRENGTHENS
1	Public interactive demo that needs no login	Judge access
2	Improved navigation and user interface	Usability
3	Unique finding ID handling to prevent duplicate identifier conflicts	Reliability
4	Clear separation of finding severity and verification status	Usability, Traceability
5	Suggested corrections and controlled draft updates	Usability
6	Fallback claim extraction when the primary AI extraction route is unavailable	Reliability
7	Improved source management and evidence traceability	Traceability
8	Persistent review decisions and stronger approval validation	Reliability
9	Improved deployment and end-to-end workflow validation	Reliability, Judge access

Usability

The public demo, clearer navigation, separate severity and verification status, and suggested corrections make findings quick to understand and act on.

Reliability

Unique finding IDs, fallback claim extraction, persistent decisions and stronger approval validation keep the workflow dependable under repeated use.

Traceability

Better source management and evidence relationships let a reviewer follow a claim to its passage and source version.

Judge accessibility

A public /demo and a validated end-to-end deployment let evaluators see the product without credentials.

15 Scope and Limitations

What Guardian AI does today, stated plainly.

✓ What it does today

- Extracts claims and verifies them against uploaded, approved sources
- Keeps humans in the approval path, enforced by the backend
- Tracks approved claims in a persistent ledger with three statuses

○ What it does not do yet

- Scheduled background monitoring and notifications
- Automatic reopening of affected reviews
- Advanced reviewer assignments and configurable approval chains
- Live monitoring of external web sources

How to read the evidence in this report

- Orbit X1 is a synthetic product, and the scenario is a demonstration, not a benchmark.
- No test counts, response times, uptime figures or accuracy percentages are claimed.
- The public demo runs on sample data; real reviews and saved decisions run behind authentication.
- Gaps are shown as Unverified, never assumed correct.

16 Future Roadmap

Planned work. None of it is implemented yet.

PHASE 1

Advanced Collaborative Workflows

- Reviewer assignments
- Finding-level role permissions
- Team activity dashboards
- Configurable approval chains
- Advanced reviewer management

PHASE 2

Scheduled Monitoring & Notifications

- Scheduled background monitoring
- Alerts when evidence changes
- Review reminders
- Advanced review reopening

PHASE 3

Advanced Evidence Intelligence

- Additional source formats
- Improved semantic retrieval
- Conflicting-source identification
- Evidence confidence indicators
- Source freshness analysis

PHASE 4

Integrations & Enterprise Readiness

- CMS & marketing integrations
- Enterprise access controls
- Audit exports
- Advanced team management
- Expanded deployment options

17 Product Value and Impact

Why this matters to the teams who publish claims.

Accuracy and trust
Helps teams identify unsupported and contradicted claims before content approval.

Evidence traceability
Connects factual claims to supporting passages and source versions.

Human control
Provides a review and approval workflow instead of letting AI approve content on its own.

Persistent knowledge
Maintains approved facts and their supporting evidence in a reusable registry.

Continuous awareness
Identifies claims that may need renewed verification when evidence changes.

Operational efficiency
Organises analysis, evidence, review, approval and claim tracking in one platform.

TARGET USERS

- Marketing teams
- Product marketing managers
- Content reviewers
- Compliance professionals
- Businesses managing product claims and factual marketing content

THE STORY IN ONE LINE

Problem, solution, product, implementation, testing, commitments kept, and a clear roadmap, all built around one idea: claims should stay connected to their evidence.

18 Evaluator's Guide

Understand Guardian AI in two minutes.

1

Open the public demo

Go to https://guardian-ai-smoky-theta.vercel.app/demo. No login is needed.

2

Explore sample findings

Open a finding, read its explanation and evidence, and press Apply or Dismiss. In the demo these only change your own screen.

3

Compare with Orbit X1

Section 11 shows the expected verification for a seven-claim draft: four supported, three contradicted, and corrections that bring all seven in line.

4

See the ledger

Section 08 explains Current & Synced, Stale Drift and Unverified, and how claims stay tied to evidence and source versions.

5

Check the engineering

Section 09 describes the architecture. The code is in the repository, and the backend runs on Render.

6

Full application

Real reviews, saved decisions and approval run behind authentication. Evaluator access is arranged through the submission form, not published here.

Two minutes, one idea. Steps 1–4 are enough to see the full loop: a claim is extracted, checked against evidence, decided by a human, and then remembered by the ledger.

19 Questions Evaluators May Ask

Straight answers.

Q Does the AI approve content?

No. Approval is a human decision, and the backend refuses approval while review decisions are pending.

Q What happens when there is no evidence for a claim?

The claim is shown as Unverified. Guardian AI does not assume a claim is correct without sufficient evidence.

Q How is this different from asking a chatbot to proofread?

Claims are extracted individually, checked by deterministic rules and against approved evidence, traced to a passage and source version, and then tracked in a persistent ledger. A one-time proofread leaves none of that behind.

Q What does Stale Drift mean?

The evidence a claim depends on has changed or become outdated, so the claim needs renewed review. Section 08 explains the ledger statuses.

Q What if the AI extraction service is unavailable?

The pipeline falls back to an alternative claim extraction route, and the deterministic rule checks are unaffected.

Q Is the public demo the real product?

It follows the same workflow on sample data and makes no backend writes. Real reviews, saved decisions and approval run behind authentication.

Q Is Orbit X1 a real product?

No. It is a synthetic test product used for validation, and the scenario is a demonstration, not a benchmark.

Q What is not built yet?

Scheduled monitoring, notifications, automatic review reopening and advanced assignments. They are listed on the roadmap in Section 16.

20 Links

Everything an evaluator needs.

RESOURCE	LINK
Project	Guardian AI: The Living Claim Ledger
Live application	https://guardian-ai-smoky-theta.vercel.app/
Public interactive demo	https://guardian-ai-smoky-theta.vercel.app/demo
GitHub repository	https://github.com/Varun-men/Guardian_AI
Backend	https://guardian-ai-1-nhbv.onrender.com

About the public demo: the demo at /demo runs on sample data and needs no login. Protected team workflows use authentication, so the public demo and the authenticated application are kept separate.



Guardian AI

The Living Claim Ledger

A claim should not simply be checked once.
It should remain connected to the evidence that supports it.

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